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Activity 4: Mentor Matching & Intervention Recommendations

Objective

The objective of this activity is to translate analytical insights derived from student scoring and segmentation into actionable mentoring decisions. This involves designing a mentor dataset, developing a mentor-student matching logic, recommending targeted interventions, and enabling alert mechanisms for high-risk students.

Input Data

The mentor matching system uses the following inputs:

- **Student Scores Dataset (students_scored.csv)**
Contains APS, WWS, PTMS, CRS, SRI, and risk classification.
- **Clustered Student Dataset (students_clustered.csv)**
Contains student segmentation labels generated using K-Means clustering.
- **Mentor Dataset (mentors.csv)**
Contains mentor profiles including expertise, specialization, availability, and experience.

Mentor Dataset Design

- The mentor dataset was created to represent a diverse pool of mentors specializing in academic guidance, wellness support, and career mentoring. Each mentor profile includes expertise area, specialization, availability, years of experience, and effectiveness rating.
- This dataset enables accurate matching between student needs and mentor expertise.

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Mentor Matching Logic

The mentor matching process follows a rule-based decision framework:

1. Student needs are identified based on performance scores and risk category.
2. High-risk students (Red category) are prioritized for wellness support.
3. Students with low academic scores are assigned academic mentors.
4. Students with low career readiness scores are assigned career mentors.
5. Remaining students receive general mentoring support.

A round-robin assignment strategy ensures fair distribution of students among mentors.

Intervention Recommendation Strategy

Each student is assigned a personalized intervention based on their identified need:

- **Academic Need:** Weekly academic reviews and structured study plans
- **Wellness Need:** Stress counseling and wellbeing sessions
- **Career Need:** Career guidance and skill roadmap development
- **General Need:** Routine mentoring and progress monitoring

These interventions are designed to address the root causes of performance gaps.

High-Risk Alert Mechanism

Students classified under the Red risk category trigger a HIGH PRIORITY alert.

This alert indicates:

- Immediate mentor engagement
- Frequent follow-ups
- Intensive support plans

The alert system ensures timely intervention to prevent academic or wellbeing deterioration.

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Final Recommendation Table

The output of the mentor matching system is the student_recommendations.csv file, which includes:

Column	Description
student_id	Unique identifier of the student
SRI	Student Readiness Index
Risk_Category	Risk classification
Primary_Need	Identified mentoring need
Assigned_Mentor_ID	Assigned mentor identifier
Assigned_Mentor_Name	Assigned mentor name
Recommendation	Suggested intervention
Alert_Level	Priority level

This table serves as the actionable decision layer for mentors and administrators.

End-to-End Workflow

1. Student data is collected and processed to compute performance scores.
2. Machine learning clustering identifies behavioral patterns and student segments.
3. Risk categories are assigned based on readiness levels.
4. Student needs are inferred using rule-based logic.
5. Mentor expertise is matched with student needs.
6. Personalized interventions are generated.
7. High-risk alerts are triggered for immediate action.

This workflow ensures that analytical insights are effectively translated into real mentoring actions.